

Lab 102: Network Traffic Analysis with Suricata

Course: Blue Team SOC Foundations

Module: IDS/IPS Deployment & Traffic Analysis

Date: June 24, 2026

Prerequisites: Lab 101 Completed, Ubuntu 24.04, GitHub Account

Lab Overview

In this lab, you will deploy Suricata, a leading open-source Intrusion Detection System (IDS), on your Ubuntu HP ProBook. You will configure it to monitor your home network traffic, generate simulated attack traffic using Nmap, and analyze the resulting alerts. This mirrors the daily workflow of a SOC Analyst monitoring SIEM dashboards.

Key Learning Objectives:

- Install and configure Suricata via CLI.
- Understand HOME_NET vs. EXTERNAL_NET variables.
- Generate and detect malicious traffic patterns (Port Scans, ICMP floods).
- Analyze JSON logs (eve.json) using jq.

Safety Note for Public/Work WiFi:

Do not run this lab on public or corporate networks. Installing IDS tools and generating scan traffic (even against yourself) can trigger corporate security alerts or violate acceptable use policies.

Wait until you are on a trusted home network where you own the router and devices.

Configuration Tip: If you are unsure of your home subnet, the default Suricata configuration often covers common private ranges (192.168.0.0/16, 10.0.0.0/8). However, explicitly defining your specific subnet reduces false positives.

Phase 1: Installation & Configuration

Step 1: Install Suricata

We will use the official OISF stable repository to ensure you have the latest version with up-to-date rule compatibility.

Task: Run the following commands in your terminal:

```
# Add the Suricata stable PPA
sudo add-apt-repository ppa:oisf/suricata-stable -y
# Update package list
sudo apt update
# Install Suricata and jq (for JSON log parsing)
sudo apt install suricata jq -y
# Verify installation
suricata --version
```

Step 2: Identify Your Network Interface

Suricata needs to know which network card to listen on.

Task: Run:

```
ip link show
```

Action: Note the name of your active interface (e.g., wlp2s0 for WiFi, eth0 for Ethernet). You will need this for the configuration.

Step 3: Configure Suricata

We need to tell Suricata what your "Home" network is and which interface to monitor.

Task: Edit the configuration file:

```
sudo nano /etc/suricata/suricata.yaml
```

Make these three changes:

1. Set HOME_NET: Find vars: -> address-groups: -> HOME_NET. Change it to match your local subnet.
 - *Common Home Networks:* [192.168.1.0/24], [192.168.0.0/24], or [10.0.0.0/8].
 - Tip: Run `ip addr` to see your IP. If you are 192.168.1.50, your network is likely 192.168.1.0/24.

```
HOME_NET: "[192.168.1.0/24]" # <--- UPDATE THIS
```

2. Set EXTERNAL_NET: Ensure this is set to !\$HOME_NET (meaning "everything else").

```
EXTERNAL_NET: "!$HOME_NET"
```

3. Set Interface: Find af-packet: and set the interface to the name you found in Step 2.

af-packet:

- interface: wlp2s0 # <--- UPDATE THIS to your interface
- cluster-id: 99
- cluster-type: cluster_flow
- defrag: yes

Save and Exit: Ctrl+O, Enter, Ctrl+X.

Step 4: Update Rules & Start Service

Download the latest threat intelligence rules and start the engine.

Task:

Update rule sources (Emerging Threats Open)

```
sudo suricata-update
```

Start Suricata

```
sudo systemctl enable --now suricata
```

Check status (Should be "active (running)")

```
sudo systemctl status suricata
```

Phase 2: Generating Attack Traffic

Now we will act as the "attacker" to trigger Suricata's detection rules.

Step 1: Monitor Alerts in Real-Time

Open a new terminal window and run this command to watch for new alerts as they happen:
`sudo tail -f /var/log/suricata/eve.json | jq 'select(.event_type=="alert") | .alert.signature'`

Note: If jq is not found, install it with `sudo apt install jq`.

Step 2: Generate Traffic (The "Attack")

Open a third terminal window and run these nmap scans against your own machine or another device on your home network (e.g., your phone or router).

Scenario A: Port Scan Detection

Replace 192.168.1.50 with YOUR IP or another device IP
`nmap -sS -T4 192.168.1.50`

Expected Alert: "ET SCAN Potential SSH Scan" or "Nmap Scripting Engine User-Agent".

Scenario B: ICMP Flood (Ping Sweep)

Send 100 rapid pings
`ping -c 100 192.168.1.1`

Expected Alert: "ICMP Ping" or "Potential ICMP Flood".

Scenario C: Malicious User-Agent Simulation

Simulate a bad bot using curl
`curl -A "Mozilla/5.0 (compatible; Nmap Scripting Engine)" http://google.com`

Phase 3: Log Analysis & Forensics

SOC analysts spend most of their time querying logs, not just watching them live.

Task 1: Query Specific Alerts

Use jq to extract high-severity alerts from the log file.

Command:

Find all alerts with "Scan" in the signatures
`sudo cat /var/log/suricata/eve.json | jq 'select(.event_type=="alert" and .alert.signature | test("Scan"))'`

Task 2: Extract Source and Destination IPs

Create a simple report of who is talking to whom.

Command:

Extract Src IP, Dest IP, and Alert Signatures
`sudo cat /var/log/suricata/eve.json | jq -r 'select(.event_type=="alert") | "Source: \(.src_ip) -> Dest: \(.dest_ip) | Alert: \(.alert.signature)'"`

Task 3: Count Alerts by Signature

Identify the most common "attacks" on your network.

Command:

`sudo cat /var/log/suricata/eve.json | jq -r 'select(.event_type=="alert") | .alert.signature' | sort | uniq -c | sort -nr | head -10`

When you have completed the lab and uploaded the materials, begin Lab 103: Incident Response with Atomic Red Team.